LASITH HAPUTHANTHRI

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PERSONAL STATEMENT

Passionate engineering student with a strong foundation in electronics and hands-on experience in designing and developing innovative solutions. I am eager to apply my skills and contribute to impactful projects in a dynamic environment.

Interested Area: Digital System Designing, Processor Designing, PCB Designing, Embedded Systems, IoT, Electronic Product Designing, Robotics, Web Development

EDUCATION

University of Moratuwa
B.Sc. Eng. (Hons.) Electronic & Telecommunication Engineering *August 2022 - Present*

Dharmaraja College Kandy *2010-2020*

- GCE Advanced Level: Physical science stream, 3A passes for Chemistry, Physics, and Combined Mathematics with a z-score of 2.23(Island Rank 329)
- GCE Ordinary Level: 9A passes

HONORS AND AWARDS

- **Champion (National Rank)| SLIOT Challenge 2023- University Category** *2023*
  HydroLink 2.0: An IoT device that can turn any domestic water tanks into smart ones.
- **Champion (National Rank)| IEEE Challenge Sphere Arduino Challege** *2024*
  REPLACE: A smart and Intelligent that is designed with custom-made analog sensors and Made to be scalable and anomaly detection.
- **Champion (National Rank) | IEEE Challenge Sphere Circuit Challege** *2024*
  StearSafe: A spectacle frame and watch that can use the EOG signals from the user to measure drowsiness and alert the user.
- **1st Runners up (National Rank) | Brainstorm The Premier Biomedical Competition** *2024*
  StearSafe: A spectacle frame and watch that can use the EOG signals from the user to measure drowsiness and alert the user.
- **Special Award for Chess**
  Dharmaraja College Kandy *2013,2016*
  Sri Lankan representation for World Youth and Asian Youth Chess Championships 2013 and got selected to World Youth and Asian Youth Chess Championships 2016.

PROJECTS

- **StearSafe -A Spectacle frame which can detect the Drowsiness state of the user and give alerts**
  Brainstorm 2024 Champion, IEEE Circuit Challenge 2024 Champion *2024*
  - A spectacle frame and smartwatch that collect EOG data, measure the drowsiness of the user and give alerts.
  - The spectacles communicate with the Raspberry Pi with an onboard Inverted F antenna.

- **HydroLink 1.0 - A Prototype to Turn Any Water Tank into a Smart One**
  EN1190: Engineering Design Project Module Project *2023*
  - An IoT device plugged into any domestic water tank checks the water level and sends it to Firebase. The water level can be monitored with a custom-made app, and the user can control the water level from the app.

- **HydroLink 2.0 - The Improved Version of HydroLink 1.0**
  UBLC World Bank Grant *Ongoing*
  - UBLC World Bank funding to produce and sell 10 devices, supporting the initiative's growth and market presence.
  - The improved version of HydroLink 1.0, featuring more compact design and minimized power consumption.

- **REPLACE: A Smart and Intelligent Wall Socket**
  IEEE Challenge Sphere Arduino Challenge *2024*
  - A smart wall socket that can detect anomalies in the domestic power grid. With a mobile app, the user can monitor power consumption and cost, and control the socket.

- **LUNA: Vision-based Automated Restaurant Robot**
  EN2160 Electronic Design Realization Module Project *2024*
  - Developing a restaurant robot using ROS and Kinect2, equipped with a Kalman-filter-based food stabilization tray to avoid beverage spillage. The robot enhances efficiency in restaurant environments with advanced sensor technologies and robotic functionalities, including a custom restaurant environment planner front-end.

- **Robot Cosmo - A Robot which can do specified tasks**
  EN2533: Robot Design and Competition Module Project *2023*
  - Tools & technologies used: Atmega2560, Mechanical Arm Design, PID, Chassis Design
  - An ATMEGA2560-based robot showcasing precise line following, wall avoidance, ramp navigation, mechanical arm interaction, sound sensitivity, color detection, maze-solving, and obstacle avoidance.

- **Robot Ratatouille - A Maze Travelling Micro-mouse Robot**
  Sri Lanka Institute of Information Technology (SLIT)-Robofest 2024 *Ongoing*
  - A fast maze-traversing micro-mouse that can navigate complex mazes autonomously using advanced sensors and algorithms.

- **Analog Function Generator**
  EN2091: Laboratory Practice and Projects Module Project *2023*
  - Tools & technologies used: LTspice, NI Multisim, Altium Designer (PCB Design), Solidworks (Enclosure Modeling)
  - An analog function generator that generates sinusoidal, triangular, sawtooth, and PWM signals up to 20kHz with variable parameters. We designed it using analog components, which feature versatile waveform options, and easy signal output.

- **IoT Medicine Storage Box**
  EN2853: Embedded Systems and Applications Project Module Project *2024*
  - Tools & technologies used: Wokwi Simulations, IoT, MQTT, Node-Red, JavaScript, ESP32
  - A medicine storage box that provides medicine time alerts, keeps the medicine in proper sunlight and monitors the temperature and humidity.
- **Single Cycle Processor**
  EN3021 Digital System Design Module *Ongoing*
  - Ongoing development of a 32-bit pipelined single-cycle RISC-V processor, implementing core modules like the Program Counter, Instruction Register, and Multiplexers in SystemVerilog. A test bench is in progress to ensure functionality and compliance with RISC-V standards.
- **Floating point Adder, Subtractor, Multiplier and Divider for IEEE 754 standards**
  EN3021 Digital System Design Module *Ongoing*
  - Ongoing development of a 32-bit floating-point Adder, Subtractor, Multiplier, and Divider following IEEE 754 standards with a comprehensive testbench to validate the accuracy and performance of all operations.
- **UART Communication Implementation Using FPGA**
  EN2111: Electronic Circuit Design *2024*
  - Tools & technologies used: Quartus Prime
  - Designed and implemented a UART communication protocol on an FPGA platform. The project involved creating a state machine to manage serial data transmission and reception between devices, utilizing both transmitter and receiver modules. The FPGA was programmed to handle baud rate synchronization and ensure accurate data exchange.
- **My Portfolio - A Website about Myself** *2024*
  - I have developed A personal portfolio website using React to showcase my projects, achievements, and PCB designs. Designed with Figma and Canva to ensure a polished, professional presentation of my work and accomplishments.

TECHNICAL SKILLS

Languages Sinhala(Native), English
Programming Python, C, C++, Matlab, Verilog, JavaScript
Soft Skills Problem Solving, Strategic Decision-Making, Project Management, Team Leadership
Developer Tools: Altium Designer, Solidworks, Geogebra, Figma, MATLAB antenna toolbox, Simulink, Github, Jupyter, Multisim, LTspice, Latex, Ultimaker Cura, Vitual Studio Code

EXPERIENCE

- **Photomath**
  Math expert *December 2021 - February 2024*
  Math Reviewer *August 2022 - February 2024*

EXTRA-CURRICULAR ACTIVITIES

Volunteering

- **Sasnaka Sansada Project Ganith Saviya** *2021-Present*
  Sasnaka Sansada Ganitha Saviya is a Sri Lankan initiative to promote and enhance mathematical education and problem-solving skills among students through competitions, workshops, and other educational activities.

- **Sri Lanka Robotics Challenge 2024** *2024*
  - Organizing Committee
  - School Workshops Presenter
  - Projects Demonstration, LUNA: Vision-based Automated Restaurant Robot
- **EXMO 2023 Technological Exhibition** *2023*
  EXMO 2023 is the flagship technological exhibition of the University of Moratuwa, Sri Lanka.
  - Projects Demonstration
    HydroLink 1.0 - A Prototype to Turn Any Water Tank into a Smart One

Sports and Games

- **Champion** in Sri Lanka All-island Youth Chess Championship 2012.
- **Champion** from team section in 4th Asian School Chess Championship 2008.
- **2nd Runners up** from the individual section in 4th Asian School Chess Championship 2008, team section in Youth Asian Chess Championship 2009
  Selected to World School Chess Championship 2016, 3-11 December in Sochi, Russia.
- Selected to 10th Asian School Chess Championship 2016 from 9-18 2016 in Tehran, **Iran** at U.12 2013 World Youth Chess Championships, Al-ain, **UAE**, at U.12 2013 Asian Youth Chess Championships, Mazandaran, **Iran**, 19-27 June 2013, Sri Lanka at the 8th Asian Schools Chess Championship 2012, New Delhi, **India** 13-20 December 2012, Sri Lanka at U.8 2009 Asian Youth Chess Championships, New Delhi, **India**, Sri Lanka at the 6th Asian Schools Chess Championship 2010, Colombo, Sri Lanka, Sri Lanka at the 5th Asian Schools Chess Championship 2009, Colombo, Sri Lanka, Sri Lanka at the 4th Asian Schools Chess Championship 2008, Colombo, Sri Lanka.
- Member of the University Table tennis pool.

Club and Organizations

- Member of the Electronic Club of University of Moratuwa *2022-Present*
- IEEE Student Member *June 2024 -Present*
- Chess Team, Dharmaraja College, Kandy, Sri Lanka. *2009-2020*
  - Captain *2013-2019*
  - Vice-Captain *2010-2012*

REFERENCES

- **Dr. Ajith A. Pasqual**
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